A Gaussian counterexample to the mixed-norm tensor identity in arXiv:2405.09378

Autonomous mathematical discovery run `fa_banach_001`

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Status. `candidate_counterexample_likely_valid`. This packet disproves the explicit identity used to motivate the source's final conjectural route; it does not solve the broader, non-theorem-shaped program of characterizing all mixed norms of metaplectic Wigner distributions.

1 Source statement

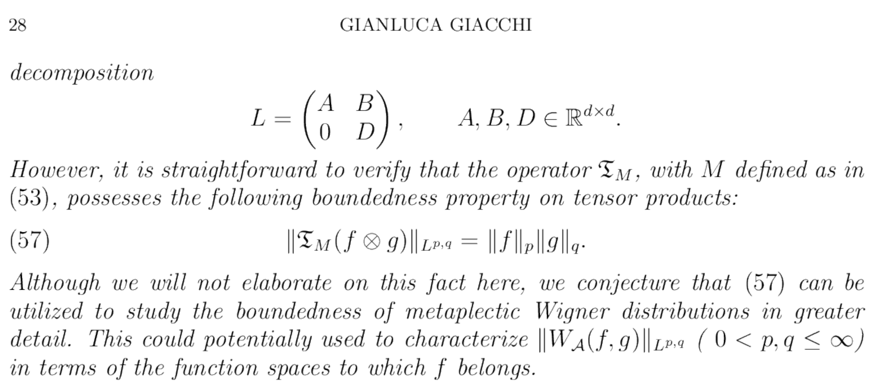
The source is Giacchi's arXiv:2405.09378, later published in the *Journal of Fourier Analysis and Applications* [1]. On page 28 of the arXiv PDF, equation (57) in the final remark states that, for

$$M = \begin{pmatrix} I + P_{12}Q_{12}^T & -P_{12} \\ -Q_{12}^T & I \end{pmatrix},$$

the rescaling operator has the tensor-product property

$$\|\mathfrak{T}_M(f \otimes g)\|_{L^{p,q}} = \|f\|_p \, \|g\|_q \,. \tag{1}$$

The remark conjectures that this identity can be used to study and possibly characterize mixed norms of metaplectic Wigner distributions. The published version contains the same assertion as equation (42) in Remark 8.5.



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decomposition

$$L = \begin{pmatrix} A & B \\ 0 & D \end{pmatrix}, \qquad A, B, D \in \mathbb{R}^{d\times d}.$$

However, it is straightforward to verify that the operator $\mathfrak{T}_M$, with M defined as in (53)*, possesses the following boundedness property on tensor products:*

$$\|\mathfrak{T}_M(f \otimes g)\|_{L^{p,q}} = \|f\|_p \|g\|_q. \tag{57}$$

Although we will not elaborate on this fact here, we conjecture that (57) *can be utilized to study the boundedness of metaplectic Wigner distributions in greater detail. This could potentially used to characterize $\|W_{\mathcal{A}}(f,g)\|_{L^{p,q}}$ ($0 < p, q \leq \infty$) in terms of the function spaces to which f belongs.*

Figure 1: The full-width source excerpt, arXiv PDF page 28.

2 Definitions and proof intuition

The source uses

$$\|F\|_{L^{p,q}} = \left(\int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} |F(x,\xi)|^p \, dx \right)^{q/p} d\xi \right)^{1/q}$$

for finite positive p, q, and $\mathfrak{T}_L F(z) = |\det L|^{1/2} F(Lz)$. We need only finite exponents.

Proof intuition. Choose $P_{12} = 0$. Then M is a determinant-one lower shear, so a tensor $f(x)g(\xi)$ becomes $f(x)g(\xi - Q^T x)$. An ordinary L^p norm cannot see such a measure-preserving shear, but a mixed norm with unequal exponents does: the inner Gaussian integration changes the covariance of the outer Gaussian. The determinant of that covariance change is exactly $\det(I + Q^T Q)$, producing a nontrivial factor whenever $p \neq q$.

3 Counterexample and exact lower-shear calculation

Theorem 1. *Let $d \geq 1$, $0 < p, q < \infty$, and $Q \in \mathbb{R}^{d \times d}$. Set*

$$P_{12} = 0, \qquad M_Q = \begin{pmatrix} I & 0 \\ -Q^T & I \end{pmatrix}, \qquad \gamma(x) = e^{-\pi|x|^2}.$$

Then, with $D = \det(I + Q^T Q)$,

$$\frac{\left\| \mathfrak{T}_{M_Q}(\gamma \otimes \gamma) \right\|_{L^{p,q}}}{\|\gamma\|_p \, \|\gamma\|_q} = D^{\frac{1}{2q} - \frac{1}{2p}}. \tag{2}$$

Consequently, if $Q \neq 0$ and $p \neq q$, the source identity (1) is false. In particular, $d = 1$, $Q = 1$, $p = 1$, and $q = 2$ give the ratio $2^{-1/4}$.

Proof. The shear has determinant one, hence

$$\mathfrak{T}_{M_Q}(\gamma \otimes \gamma)(x,\xi) = \exp\bigl(-\pi(|x|^2 + |\xi - Q^T x|^2)\bigr).$$

Put $A = I + QQ^T$. Completing squares gives

$$\begin{aligned} |x|^2 + |\xi - Q^T x|^2 = {} & (x - A^{-1}Q\xi)^T A (x - A^{-1}Q\xi) \\ & + \xi^T (I + Q^T Q)^{-1} \xi. \end{aligned}$$

Here we used the standard identities $I - Q^T A^{-1} Q = (I + Q^T Q)^{-1}$ and $\det A = \det(I + Q^T Q) = D$. Gaussian integration in x therefore yields

$$\left\| \mathfrak{T}_{M_Q}(\gamma \otimes \gamma)(\cdot, \xi) \right\|_p = p^{-d/(2p)} D^{-1/(2p)} \exp\bigl(-\pi \xi^T (I + Q^T Q)^{-1} \xi\bigr).$$

Taking the outer L^q norm gives

$$\left\| \mathfrak{T}_{M_Q}(\gamma \otimes \gamma) \right\|_{L^{p,q}} = p^{-d/(2p)} q^{-d/(2q)} D^{-1/(2p) + 1/(2q)}.$$

On the other hand, $\|\gamma\|_p \, \|\gamma\|_q = p^{-d/(2p)} q^{-d/(2q)}$, proving (2). If $Q \neq 0$, the positive semidefinite matrix $Q^T Q$ is nonzero, so $D > 1$; unequal exponents then make the last power nonzero. $\square$

Remark 1 (Exact correction within the chosen subfamily)**.** *For $P_{12} = 0$, identity (1) holds for all tensor inputs when $p = q$, because $L^{p,p} = L^p$ and M_Q is measure preserving. It also holds when $Q = 0$, when $M_Q = I$. Theorem 1 proves that for $0 < p, q < \infty$, these are the only possibilities if the identity is required for every tensor input.*

4 Verification, scope, and novelty

The proof is exact and uses no numerical approximation. A reviewer should check two convention-sensitive points: the source puts the L^p norm in the x variable inside the L^q norm in ξ, and the determinant factor in $\mathfrak{T}_M$ is one for M_Q. Both match the source definitions.

A bounded search on 11 August 2026 used the exact equation phrase, the title, arXiv id 2405.09378, "mixed norm tensor products," and "erratum." It found the arXiv and final journal versions, both containing the assertion, but no erratum or prior correction. Novelty confidence is moderate because this is an elementary Gaussian calculation that may be informally known.

The result does not characterize $\|W_{\mathcal{A}}(f, g)\|_{L^{p,q}}$ in general. It shows that the proposed route cannot begin from (1) without additional hypotheses or a replacement inequality.

Human-review recommendation. Accept as a likely-valid counterexample to equation (57)/(42), after checking the mixed-norm order. Do not count it as a full solution of the broad final research program.

References

[1] G. Giacchi, *Boundedness of metaplectic operators within L^p spaces, applications to pseudodifferential calculus, and time-frequency representations*, J. Fourier Anal. Appl. **30** (2024), Article 69; arXiv:2405.09378, https://arxiv.org/abs/2405.09378.